

A packing problem

Suppose you have a collection of 32 GB USB memory sticks. You need to copy 20 files of different sizes onto these memory sticks. The file sizes (in GB) are:

9.6	3.1	13.8	8	13.5	3.5	16.3	7	12.2	3.8
5.2	15.9	2.7	14.3	16.2	1.3	7	12.0	8.2	16.4

What is the fewest number of memory sticks that you will need to use?

Chris has 200 music and movie files that he needs to transfer to his digital collection. The file sizes (in GB) are shown below. He will store these on hard disk drives which have a capacity of 500 GB. How should he allocate the files to the hard drives?

5.0	4.5	5.1	98.0	2.4	174.7	3.0	180.6	140.2	2.9
4.1	3.6	4.7	6.7	4.5	138.9	6.0	97.3	129.1	4.3
5.0	97.1	5.1	121.0	4.1	115.2	3.0	4.3	3.2	6.3
2.4	5.7	6.9	4.7	135.9	3.8	3.9	103.9	3.9	6.0
2.5	6.5	4.6	3.2	4.1	6.5	2.4	2.9	4.8	136.3
2.3	6.3	6.7	5.3	4.4	144.7	2.7	5.4	6.1	5.4
2.2	3.4	2.3	2.0	4.3	6.5	4.7	4.7	6.2	2.7
2.5	6.4	4.5	6.2	6.8	2.4	4.2	149.8	6.4	2.1
3.4	166.4	4.5	5.6	4.8	93.1	3.1	5.5	4.1	3.8
5.2	4.9	4.2	138.9	174.9	6.8	6.1	2.9	132.8	4.3
6.8	96.2	3.3	4.0	3.7	4.4	136.0	6.1	2.7	179.9
5.0	5.9	4.5	174.0	3.2	173.8	6.4	116.3	3.4	5.0
153.6	127.3	112.1	4.6	100.6	4.1	3.2	151.7	174.6	149.5
165.3	3.9	5.3	6.0	2.2	5.0	105.7	3.3	5.0	6.0
3.9	170.3	5.5	3.5	2.9	3.1	6.0	101.4	3.6	4.0
3.9	3.1	161.0	150.1	5.1	3.1	5.1	2.2	5.1	6.6
173.1	4.9	6.7	4.4	140.0	2.6	105.4	140.7	2.1	4.6
139.6	3.8	135.5	3.3	117.2	137.6	3.9	3.8	4.3	3.5
3.8	6.3	5.9	4.4	3.1	5.0	5.1	100.5	3.1	3.3
2.6	4.8	3.4	2.8	4.3	7.0	149.8	5.4	91.0	5.1